The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 September 1988.

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR,
N-CHANNEL, SILICON LOGIC-LEVEL,
TYPES 2N6902 AND 2N6904
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a logic-level N-Channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1. TO-204AA for 2N6902, 2N6904 formerly TO3.

1.3 Maximum ratings.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} T_C = +100°C	I_S 2/	I_{DM}	T_{op} and T_{stg}
	W	W	V dc	V dc	V dc ±10	A dc	A dc	A dc	A(pk)	°C -55, +150
2N6902	75	4	100	100		12	7.6	12	30	
2N6904	75	4	200	200		8.0	5.1	8.0	20	

1/ Derate linearly T_C > +25°C - 2N6902, 2N6904 (0.6 W/°C).

2/ Derate above T_C = +25°C according to the formula $I_D = \sqrt{\frac{P(rated)}{K}}$, where P(rated) = P_T-(T-25) (W/°C) watts; K = Max $r_{DS(on)}$ at T_J = +150°C.

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 1 mA	Max I_{DSS1} V_{GS} = 0 V		$R_{\theta JC}$ maximum	Max $r_{DS(on)}$ 1/ V_{GS} = 5 V dc	
			V_{DS} = 80 V	V_{DS} = 160 V		T_J = +25°C at I_{D1}	T_J = +150°C at I_{D2}
	V dc	V dc Min 1.0 Max 2.0	μA dc		°C/W	Ohms	Ohms
2N6902	100		1.0	---	1.67	0.2	0.52
2N6904	200		---	1.0	1.67	0.6	1.17

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

$V_{(BR)DSS}$ - Drain to source breakdown voltage, all other terminals short-circuited to source.
I_S - Source current through drain diode (forward baised V_{SD}).
$Q_{g(th)}$ - Minimum gate charge that must be supplied before the device will turn on.
$Q_{g(on)}$ - Gate charge that must be supplied to ensure that the device is in the on-state.
$Q_{gm(on)}$ - Gate charge that must be supplied to the device to reach the maximum rated gate-source voltage.
C - Coulomb.
g_{FS} - DC forward transconductance.
$V_{GS(on)}$ - Gate voltage required to reach the specified drain current.
Q_{gs} - Charge required to reach the gate charge test current.
Q_{gd} - Charge supplied to the gate while the drain waveform is switching.
$I_{(ISO)}$ - Source pin-to-case isolation current.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

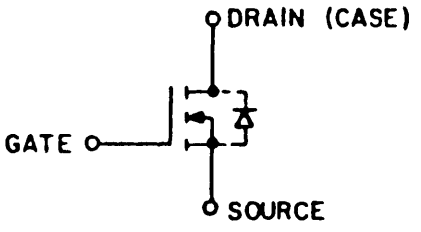
3.3.1 Lead material and finish. Lead material shall be Kovar, Alloy 52; a copper core is permitted. Lead finish shall be gold, tin, or solder. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.3).

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

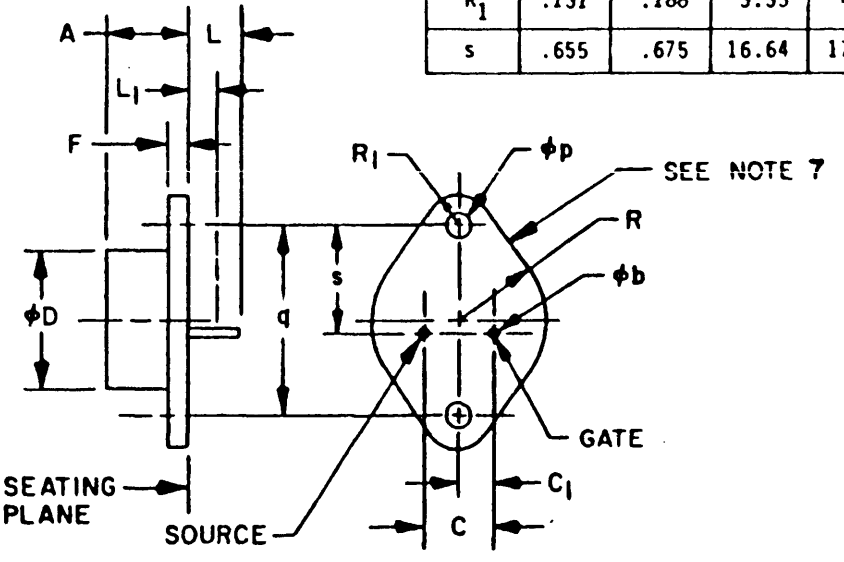
3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection (see 6.2).

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SCHEMATIC CIRCUIT

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.360	6.35	9.15	1
øb	.038	.043	0.97	1.10	
øD	---	.875	---	22.23	
C	.420	.440	10.67	11.18	1,6
C_1	.205	.225	5.21	5.72	1,6
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1	---	.050	---	1.27	
øp	.151	.161	3.84	4.09	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	



NOTES:
1. These dimensions should be measured at points .050 (1.27 mm), .055 (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
2. The seating plane of the header shall be flat within .001 (0.03 mm) concave to .004 (0.10 mm) convex inside a .930 (23.62 mm) diameter circle on the center of the header and flat within .001 (0.03 mm) concave to .006 (0.15 mm) convex overall.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. Mounting holes shall be deburred on the seating plane side.
6. These dimensions pertain to the 2N6902 and 2N6904 types.
7. Drain is electrically connected to case.

FIGURE 1. Physical dimensions for TO-204AA.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.3 Screening (JANTX, JANTXV and JANS levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	Test condition F, except T_{low} = -55°C, and 20 cycles	Test condition F, except T_{low} = -55°C, and 20 cycles
1/ 2/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9	I_{GSS1}, I_{DSS1}, gate stress test (see 4.5.5) Subgroup 2 of table I herein	Gate stress test (see 4.5.5) Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	Subgroup 2 of table I herein I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±.2 µA dc or ±100% of initial value, whichever is greater.	Subgroup 2 of table I herein I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)}$
12	Method 1042, test condition A and test condition C (see 4.3.1)	Method 1042, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±.2 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20% of initial value. $\Delta V_{GS(th)1}$ = ±20% of initial value.	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±.2 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20% of initial value. $\Delta V_{GS(th)1}$ = ±20% of initial value.

1/ Shall be performed anytime before screen 9.

2/ This test method in no way implies an avalanche energy rating. This is a stress test designed to ensure a rugged product.

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4.3.1 Power burn-in. Power burn-in conditions are as follows: T_A = +25°C -5°C, +10°C; $V_{DS} \geq 10$ V minimum; I_D adjusted to meet a junction temperature of +140°C -0°C, +10°C by controlling V_{GS} voltage to obtain a specific I_D current required to apply an appropriate power for the device under stress. Power condition and T_J requirements can be established by ΔV_{SD} measurement technique described in MIL-STD-750, method 3161. For condition C, t = 240 hours.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and table IIa and table IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal impedance. Thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC(max)}$ = .6°C/W.

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	2 A.
c.	t_H heating time	Steady-state (see MIL-STD-750 method 3161 for definition).
d.	V_H drain-source heating voltage	25 V.
e.	t_{MD} measurement time delay	10 to 80 µs.
f.	t_{SW} sample window time	10 µs maximum.

4.5.3 Thermal response (ΔV_{SD} measurements). The delta V_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	2 A minimum.
c.	t_H heating time	100 ms.
d.	V_H drain-source heating voltage	25 V minimum.
e.	t_{MD} measurement time delay	10 to 80 µs.
f.	t_{SW} sample window time	10 µs maximum.

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4.5.4 Unclamped inductive switching.

a. Peak current, I_D - - - - - - - - - 3.0 A.

b. Peak gate voltage, V_{GS} - - - - - - 10 V.

c. Gate to source resistor, R_{GS}

d. Initial case temperature - - - - - - +25°C +10°C, -5°C.

e. Inductance, L- - - - - - - - - - - - 100 µH ±10%.

f. Number of pulses to be applied - - - 1 pulse.

4.5.5 Gate stress test.

V_{GS} = 15 V minimum.
t = 250 µs minimum.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R $\leq$ 100 K, whenever bias voltage is to be applied drain to source.

6.3 Ordering data. Acquisition documents may specify the lead finish if other than specified (see 3.3.1).

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071							
Subgroup 2								
Breakdown voltage drain to source	3407	I_D = 1.0 mA dc, Bias condition C, V_{GS} = 0 V			$V_{(BR)DSS}$			
2N6902						100	---	V dc
2N6904						200	---	V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1.0 mA dc			$V_{GS(th)1}$	1.0	2.0	V dc
Gate current	3411	V_{GS} = ±10 V dc, Bias condition C, V_{DS} = 0 V			I_{GSS1}	---	±100	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V			I_{DSS1}			
2N6902		V_{DS} = 80 V dc				---	1.0	µA dc
2N6904		V_{DS} = 160 V dc				---	1.0	µA dc
Static drain to source "on-state" resistance	3421	V_{GS} = 5 V dc; Bias condition A; Pulsed (see 4.5.1)			$r_{DS(on)1}$			
2N6902		I_D = 7.6 A dc				---	0.2	ohm
2N6904		I_D = 5.1 A dc				---	0.6	ohm
Drain to source "on-state" voltage	3405	V_{GS} = 5 V dc; Bias condition A; Pulsed (see 4.5.1)			$V_{DS(on)}$			
2N6902		I_D = 12.0 A dc				---	3.3	V
2N6904		I_D = 8.0 A dc				---	5.5	V
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V			V_{SD}			
2N6902		I_S = 12.0 A dc				0.8	1.6	V
2N6904		I_S = 8.0 A dc				0.8	1.6	V
Forward transconductance	3475	Pulsed (see 4.5.1)			g_{FS}			
2N6902						3	12	S
2N6904						3	12	S

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 3								
High temperature operation:		$T_C = T_J = +125°C$						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = 1.0$ mA dc			$V_{GS(th)2}$	0.5	---	V dc
Gate current	3411	$V_{GS} = \pm 10$ V dc, Bias condition C, $V_{DS} = 0$ V			I_{GSS2}	---	±200	nA dc
Drain current	3413	Bias condition C, $V_{GS} = 0$ V			I_{DSS2}			
2N6902		$V_{DS} = 80$ V dc				---	50	µA dc
2N6904		$V_{DS} = 160$ V dc				---	50	µA dc
Static drain to source "on-state" resistance	3421	$V_{GS} = 5$ V dc Pulsed (see 4.5.1)			$r_{DS(on)2}$			
2N6902		$I_D = 7.6$ A dc				---	0.32	ohm
2N6904		$I_D = 5.1$ A dc				---	1.11	ohm
Low temperature operation:		$T_C = T_J = -55°C$						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = 1.0$ mA dc			$V_{GS(th)3}$	---	3.0	V dc
Subgroup 4								
Switching time test	3472	I_D = half rated I_{D1}, $R_{GEN} = 15\Omega$, $R_{GS} = 15\Omega$, $V_{GS} = 5$ V, V_{DD} = 50% of rated V_{DD}						
Turn-on delay time					$t_{d(on)}$			
2N6902						---	50	ns
2N6904						---	45	ns
Rise time					t_r			
2N6902						---	150	ns
2N6904						---	150	ns
Turn-off delay time					$t_{d(off)}$			
2N6902						---	130	ns
2N6904						---	135	ns
Fall time					t_f			
2N6902						---	150	ns
2N6904						---	105	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 5								
Safe operating area	3474	See figure 3						
High voltage test		V_{DS} = 80% of rated V_{DS}						
2N6902								
2N6904								
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7						
Subgroup 6								
Not applicable								
Subgroup 7								
Gate charge	3471	Condition A or B						
Minimum off-state gate charge					$Q_{g(th)}$			
2N6902						0.5	1.0	
2N6904						0.5	0.9	
On-state gate charge					$Q_{g(on)}$			nC
2N6902						11	21	
2N6904						10	20	
Maximum on-state gate charge					$Q_{gm(on)}$			nC
2N6902						21	39	
2N6904						21	39	
Gate plateau voltage					V_{GP}			
2N6902						2.7	5.0	
2N6904						2.5	4.7	
Gate to source charge					Q_{gs}			nC
2N6902						1.1	5.0	
2N6904						1.4	5.0	
Gate to source charge					Q_{gd}			nC
2N6902						5	16	
2N6904						3	14	
Reverse recovery time	3473	$V_{DD} \leq 30$ V			t_{rr}			
2N6902		di/dt = 100 A/µs I_F = 4 A				---	375	ns
2N6904		di/dt = 100 A/µs I_F = 4 A				---	625	ns

1/ LTPD numbers are to be taken from MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051	Test condition F3, except T_{low} = -55°C		
Hermetic seal a. Fine leak b. Gross leak	1071			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		
Decap internal visual design verification	2075	See 3.3.2		
SEM	2077			
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately.		
Die shear strength	2017			
Subgroup 4				
Intermittent operation life	1042	Condition D (see 4.3.1), power-on time shall be 3 minutes minimum 2,000 cycles. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle.		
For 2N6902 and 2N6904		V_{DS} = 20 V dc; P_T = 4 W at T_A = +25°C ±3°C		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8		

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5				
Accelerated steady-state operation life	1042	120 hours; see 4.3.1 T_J = +200°C. Marking legibility requirements shall not apply.		
For 2N6902 and 2N6904		V_{DS} = 20 V dc; P_T = 4 W at T_A = +75°C ±3°C		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		
Bond strength (Al-Au die interconnects only)	2037	Test condition A		
Subgroup 6				
Thermal impedance	3161	See 4.5.3		

1/ LTPD numbers are to be taken from MIL-S-19500.

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TABLE IIb. Group B inspection for JANTX and JANTXV.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051	Test condition F1, except T_{low} = -55°C and 25 cycles	
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			
Intermittent operation life (LTPD)	1042	Test condition D, 2,000 cycles. A cycle shall be 1 minute minimum.	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7 and 8	
Subgroup 4			
Decap internal visual design verification	2075	See 3.3.2	
Bond strength	2037	Test condition A, all internal bond wires for each device shall be pulled separately.	
Subgroup 5			
Qualification only			
Thermal impedance	3161	See 4.5.2	
Subgroup 6			
High-temperature (nonoperating) life (LTPD)	1032	T_A = +150°C	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	

1/ LTPD numbers are to be taken from MIL-S-19500.

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TABLE III. Group C inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	1/ LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 1							
Physical dimensions	2066	See figure 1					
Subgroup 2							
Thermal shock (glass strain)	1056						
Hermetic seal	1071						
a. Fine leak b. Gross leak							
Moisture resistance	1021						
Visual and mechanical evaluation	2071						
Electrical measurement		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					
Subgroup 3							
Shock	2016						
Vibration, variable frequency	2056						
Constant acceleration	2006						
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					
Subgroup 4							
Salt atmosphere (corrosion)	1041						
Subgroup 5							
Not applicable							
Subgroup 6			3				
Intermittent operation life (LTPD)	1042	Test condition D, 6,000 cycles. A cycle shall be 1 minute minimum.					
Hermetic seal	1071						
a. Fine leak b. Gross leak							
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8					

See footnote at end of table.

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TABLE III. Group C inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	1/ LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 7			3				
Qualification only							
Thermal shock temperature cycling destructive	1051	-55°C to 150°C 200 cycles					
Hermetic seal a. Fine leak b. Gross leak							
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8					
Subgroup 8			10				
Qualification only							
Steady-state reverse bias	1042	Test condition A, 1,000 hours					
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					
Subgroup 9			10				
Qualification only							
Steady-state gate bias	1042	Test condition B, 1,000 hours V_{GS} = rated voltage					
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					

1/ LTPD numbers are to be taken from MIL-S-19500.

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TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Breakdown voltage, drain to source	3407	I_D = 1 mA dc, Bias condition C, V_{GS} = 0 V	$V_{(BR)DSS}$			
	2N6902				100	---	V dc
	2N6904				200	---	V dc
2.	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)1}$	1.0	2.0	V dc
3.	Gate current	3411	V_{GS} = ±10 V dc Bias condition C, V_{DS} = 0 V	I_{GSS1}	---	±100	nA dc
4.	Drain current	3413	Bias condition C, V_{GS} = 0 V	I_{DSS1}			
	2N6902		V_{DS} = 80 V dc		---	50	µA dc
	2N6904		V_{DS} = 160 V dc		---	50	µA dc
5.	Static drain to source "on-state" resistance	3421	V_{GS} = 5 V dc, Bias condition A, Pulsed (see 4.5.1)	$r_{DS(on)1}$			
	2N6902		I_D = 7.6 A dc		---	0.2	Ohm
	2N6904		I_D = 5.1 A dc		---	0.6	Ohm
6.	Drain to source "on-state" voltage	3405	V_{GS} = 5 V dc, Bias condition B, Pulsed (see 4.5.1)	$V_{DS(on)}$			
	2N6902		I_D = 12.0 A dc		---	3.3	V
	2N6904		I_D = 8.0 A dc		---	5.5	V
7.	Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0	V_{SD}			
	2N6902		I_S = 12.0 A dc		0.8	1.6	V
	2N6904		I_S = 8.0 A dc		0.8	1.6	V
8.	Thermal response	3161	See 4.5.2	ΔV_{SD}	1/		
	2N6902				---		
	2N6904				---		

1/ Ten percent degradation in group B is permitted.
Thirty percent degradation in group C is permitted.

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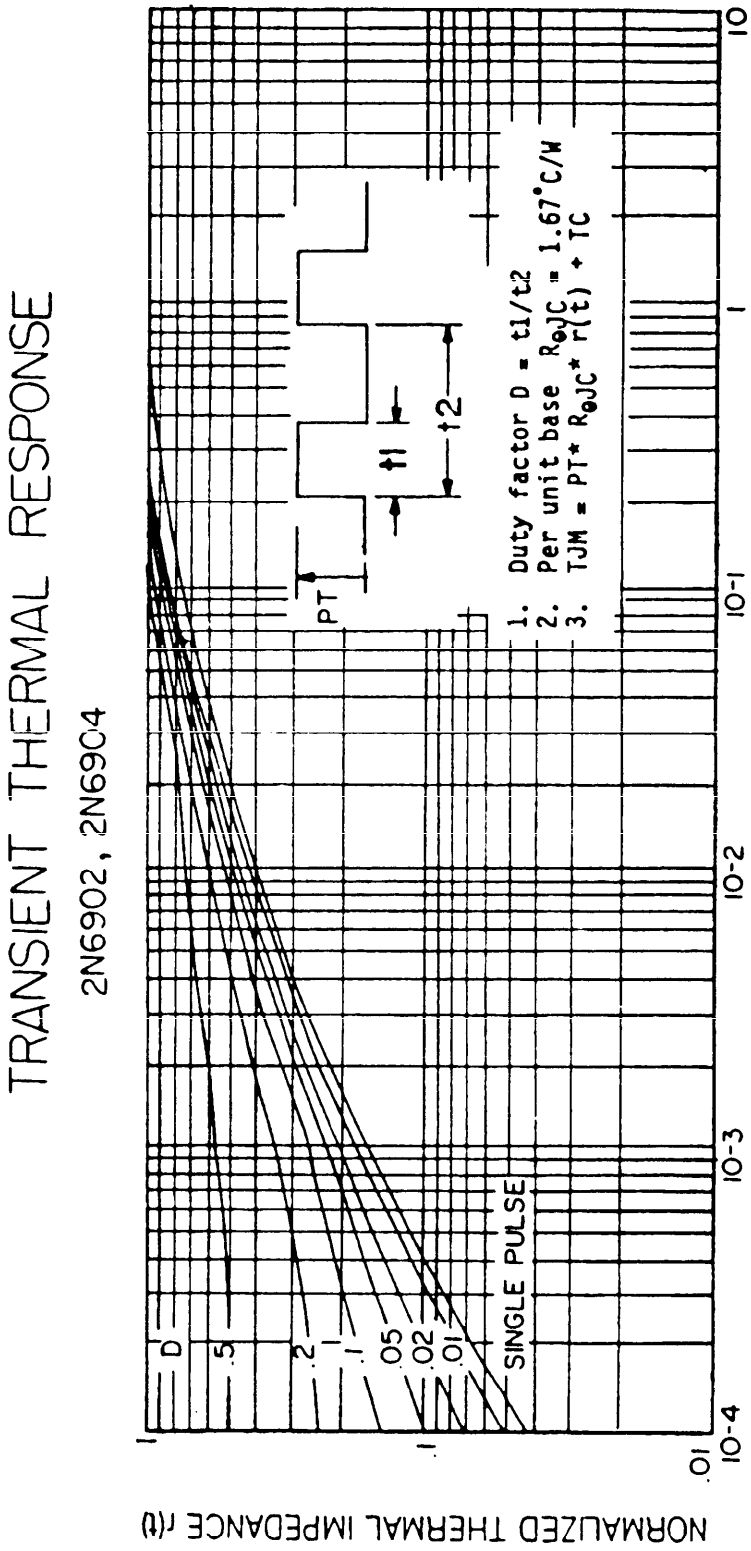


FIGURE 2. Transient thermal response.

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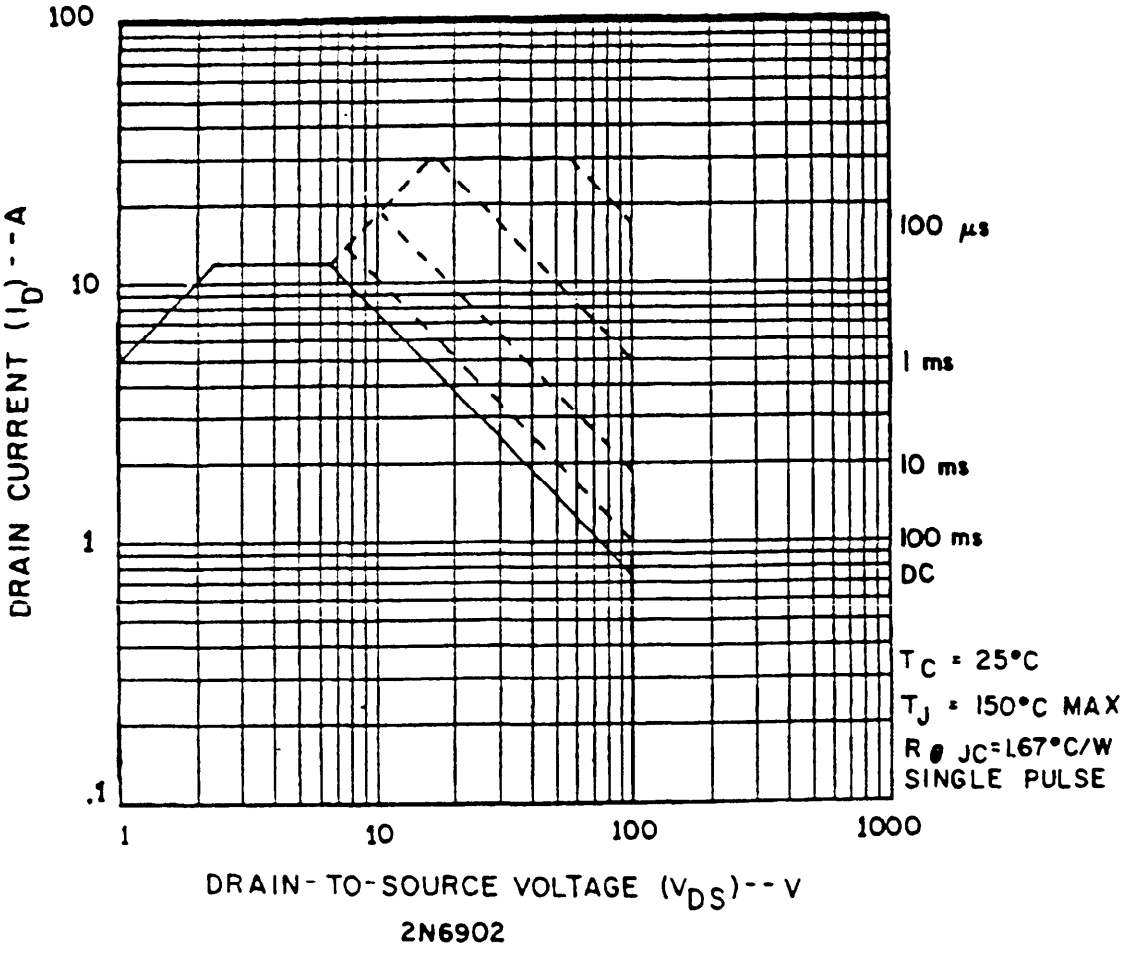


FIGURE 3. Maximum safe operating area.

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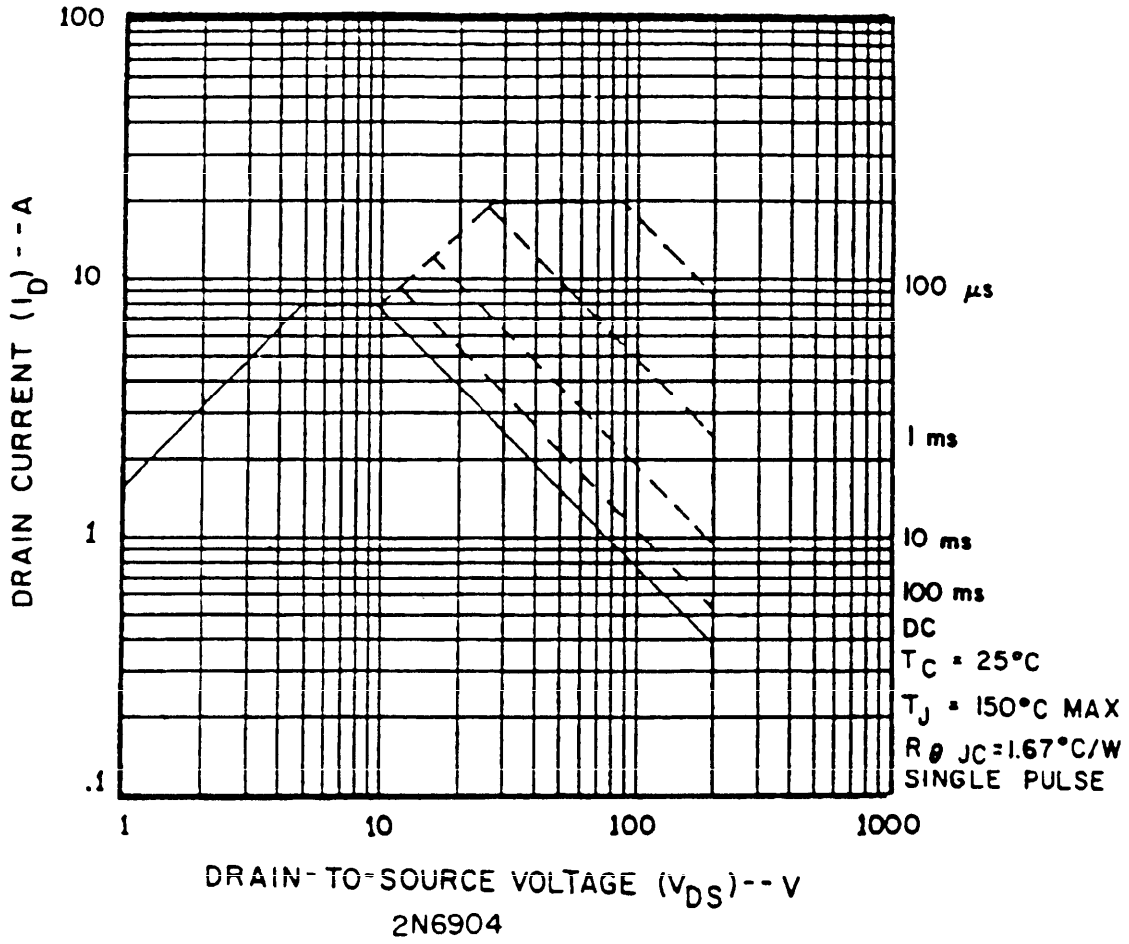


FIGURE 3. Maximum safe operating area - Continued.

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES
NASA - EB13

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13, 15

Preparing activity:
Army - ER

Agent:
DLA - ES

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